

CRL 5X / ORIGINAL CONFIGURATION

Broker margin, risk and capacity.

The original configuration leaves positive modelled margin on average in all three books. This margin must cover acquisition, additional support, the actual provider price and capital: it is not yet profit after all costs. Aggregate risk measures do not confirm uniformly lower broker risk.

18 correlated variants, 1,000 clients with \$50,000 initial capital, a 24-month horizon. No additional assumptions about premium clients. Provider at theoretical premium Q; A-book execution at 5 seconds + 0.5 pip. The three book models remain distinct.

Book	Average margin \$ / client	Min / max \$	Positive
A-book	2,127	1,725 / 2,449	18/18
Hybrid 10% B-book	2,569	1,845 / 3,291	18/18
Pure B-book	6,542	-283 / 12,467	15/18

Margin available for additional costs

Residual margin = modelled margin - CAC -24 × additional monthly support -2 × annual cost of capital × allocated capital -provider loading × Q × external share. Rates are fractions. Capital is allocated throughout the period; the model does not certify its amount. Zeros show margin before additional costs, not the parity threshold with traditional leverage.

Arithmetic example, not a quote: average cumulative Q is \$95,688 per client over 24 months. A 1% provider loading uses \$957 of the \$2,127 A-book margin, leaving \$1,170 before other missing costs. This is below the traditional \$1,555: parity is lost at 1%. Average parity is about 0.60% above Q (0.35-0.89% across variants), not zero CRL margin.

An additional offering and opportunity cost

Incremental contribution deducts foregone traditional-leverage margin on the migrated-client share from CRL margin. Acquisition costs apply to new clients; transition costs to migrated clients. This separates new business and migration without requiring CRL to replace all traditional leverage. A homogeneous mix is assumed: the model does not estimate how many clients will buy CRL.

Economic conclusion

Modelled margin must cover additional costs and be checked with the broker and provider. A-book and the 10% hybrid have positive margin in 18/18 variants at modelled costs; B-book in 15/18. Positive margin does not mean margin above traditional leverage. Unknown costs prevent a claim of full profit or a risk-adjusted return.

PORTFOLIO OF 1,000 CLIENTS / REALISED CASH FLOWS

Risk does not decrease in every dimension.

In the CRL B-book, margin is positive in 15/18 variants. The worst loss over a complete day is \$621,961 versus \$1,411,410 for traditional. Maximum cumulative decline from peak, however, is \$5,513,276 versus \$4,333,381. These are maxima along simulated historical portfolio paths of 1,000 clients, not quantiles or regulatory capital.

Book / product	Worst day \$	Maximum drawdown \$	Maximum initial deficit \$
A-book / STD	4,167	4,167	0
A-book / CRL	91,236	102,212	10,927
Hybrid 10% B-book / STD	132,981	315,338	51,148
Hybrid 10% B-book / CRL	115,233	199,737	148,381
Pure B-book / STD	1,411,410	4,333,381	794,816
Pure B-book / CRL	621,961	5,513,276	3,675,386

These measures are maxima across correlated variants and may occur at different times. They must not be added together. Incomplete UTC days at the endpoints are excluded from the daily metric. The deficit includes all modelled payment dates, including intraday dates.

Interpretation

The client result is a B-book liability. A smaller maximum daily loss does not imply a more stable cumulative balance. The initial deficit is the broker's own cash that would have prevented a negative balance on these assumed flows, starting from zero and retaining every profit. It does not measure collateral, regulatory capital or mark-to-market losses on open positions.

The recognition convention matters

Commissions, premium, funding and payoff are recognised at closure as in the baseline; execution adjustments at the fill. The fixed cost of \$100 per client over 24 months is allocated monthly. Advance premiums, segregation, collateral or different settlement dates may alter actual treasury requirements. Client deposits are not treated as available broker equity.

OPERATIONAL SENSITIVITIES / NO ASSUMED CAPITAL REQUIREMENT

Trigger concentration. Liquidity and execution.

Assumed provider delay	Peak A-book 4x payout advance \$	Hybrid 10% \$
1 hour	660,614	594,553
5 hours	889,472	800,525
24 hours	1,370,630	1,233,567

Positive incremental payout only, with certain reimbursement after the delay: not total funding needs. Excludes return of deposits, the underlying leg, margins, prefunding and default. Pure B-book has no such external reimbursement item. The maximum observed activation count is 904 in 5 minutes and 1,300 in 1 hour; the same client may trigger more than once.

Additional adverse slippage	A-book 1h loss \$	Hybrid 10% \$
1 pip	27,075	24,368
5 pip	135,375	121,838
10 pip	270,751	243,676
25 pip	676,877	609,189

Static sensitivity at the maximum historical concentration of 5x notional at the trigger, converted at entry price. No new trigger or future path, and no probability assigned. Client payment is crystallised: execution delay is borne by the broker on the external share.

Reproducibility and limitations

36 ledgers, 18 pairs, 4 stress cases, 3 books. 10,368 monthly reconciliations, error below \$0.000001. Time windows verified using direct sums and boundary tests; input hashes retained. No new client simulation. Internal checks, not a new independent review. Variant-level and monthly data are in the ZIP.

To complete a commercial assessment

The broker_inputs_template.json form distinguishes unknown values from zero: CAC by segment, support, executable provider loading, capital and its cost, migrated share, transition, collateral and payment timing. Product understanding, willingness to pay and repeated use require testing with real clients. They cannot be inferred from the premium label.